

LSTM Forecasting Workflow

Load data (xlsx) → df

Clean & tidy
fill(YEAR),
Month → Month_num,
date

Sort by date
select(date, prevalence)

Difference series
prevalence → diffed
(first NA → 0)

Lag transform → supervised
(x-1, x)

Time-ordered split
70% Train
30% Test

Scale to [-1, 1]
(train min/max only)

Reshape to 3D
[samples,
timesteps = 1,
features = 1]

Build LSTM Model
(Stateful) + Dense(1)

Train Model
shuffle = FALSE
reset_states()

Predict
Train & Test

Invert Scaling
→ Original diff units

Invert Differencing
 $\hat{Y}_{level} = \hat{Y}_{diff} +$
Previous Actual

Evaluate
RMSE / MAE

Visualize Results
Train vs Test
(ggplot2)